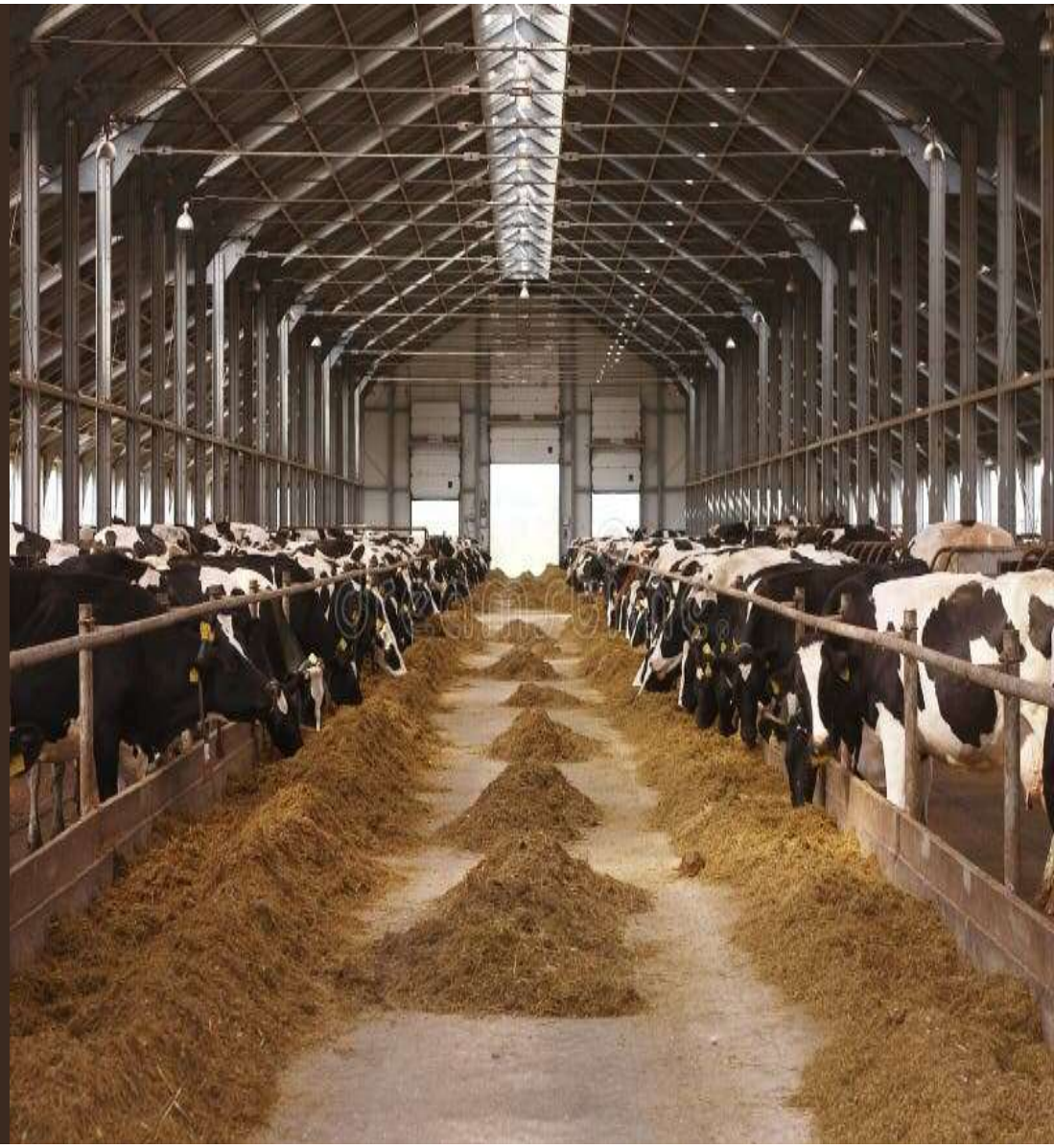
AI-Based Non-Invasive Cow Feeding and Rumination Monitoring

Using computer vision and deep learning for early-warning detection of livestock health problems



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Illness is spotted only after the milk drops

On most dairy farms, a sick cow is identified by the financial damage she causes, not by the earliest behavioral changes she shows.

STAGE ONE

Incubation

Illness is developing but is invisible to staff

STAGE TWO

Behavior drops

Feeding and rumination decline, unnoticed between vet visits

STAGE THREE

Milk crashes

Yield falls, treatment starts, and the loss is already permanent

A lagging indicator

Metabolic and infectious illnesses are typically noticed only after milk yield has collapsed. By then the animal has been unwell for days and recovery is slower.

A veterinary blind spot

Vets rely on periodic physical visits. Between inspections there is no continuous, objective record of how each cow is eating and ruminating.

A costly delay

Late clinical intervention increases antibiotic use and treatment cost, reduces animal welfare, and causes permanent loss of milk revenue.

Zero-touch monitoring: the camera replaces the collar

Existing automated tracking depends on devices attached to the animal. Our approach moves the work into software and uses video the farm already records.

CURRENT APPROACH

Wearable sensors

- Collars, ear tags and pedometers can break, slip off or lose signal in the barn
- High upfront capital cost, multiplied across every animal in the herd
- Fitting and replacing devices requires handling, which stresses the cows
- Batteries, repairs and re-tagging create a constant maintenance burden for staff

OUR APPROACH

Overhead camera video

- Works with the overhead barn cameras that farms already have installed
- Software-first: no hardware is attached to any animal
- No handling or fitting, so no added stress on the herd
- Maintenance is limited to the camera and the software, not every cow



Result: the hardware burden shifts to infrastructure the farm already owns, with zero stress and zero maintenance for the animals.

From barn video to daily behavior data in four steps

A video-upload software layer turns raw footage into a structured daily record for every individual cow.

01

Video upload

Staff upload a daily batch of footage (.mp4 or .avi) from existing barn security cameras through a simple dashboard. No new hardware is required.

INPUT: .MP4 / .AVI

02

Detect, track and identify

YOLO detects each cow and DeepSORT follows her across frames. Coat-spot patterns or OCR of visible ear tags re-identify her every day.

YOLO · DEEPSORT · OCR

03

Segment behavior

Deep learning layers label every frame as feeding, ruminating or other, converting raw video into precise daily time budgets.

DEEP LEARNING

04

Store and score

Per-cow results are saved to the database and passed to the scoring engine, which compares each cow with her own baseline.

DATABASE · SCORING

Output: one daily feeding and rumination time budget for every cow, ready for baseline comparison and risk scoring.

Two behaviors, two computer-vision strategies

Feeding is a spatial problem and is highly reliable. Rumination is a micro-movement problem that needs a dedicated jaw-focused pipeline.

HIGH RELIABILITY

Feeding duration

- *A virtual bounding box is drawn over the feed bunk in the camera view to define the feeding zone.*
- *A head-down posture inside the zone starts a duration counter that stops when she steps away.*
- *Total daily time past the feeding threshold is logged against her ID.*

FEED BUNK ZONE

Head-down posture inside this zone counts as feeding time

MICRO-MOVEMENT CHALLENGE

Rumination detection

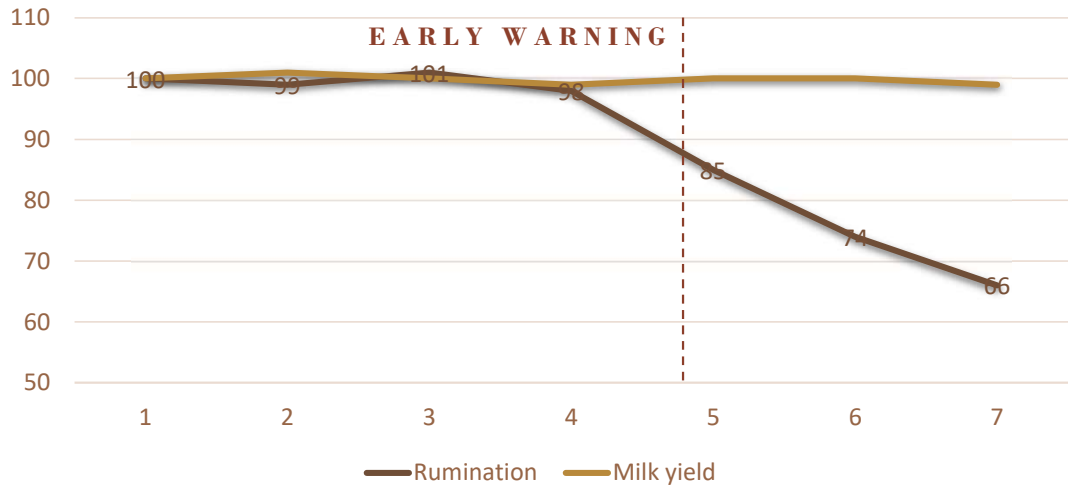
- *A multi-stage deep learning model locks onto a dynamic region of interest around the lower jaw and muzzle.*
- *Optical flow or an LSTM temporal network measures the rhythmic chewing loop frame by frame.*
- *A sustained chewing cycle confirms active rumination, and the time is logged.*



Illustrative jaw-motion signal: a steady, rhythmic chewing cycle

Catching illness before the milk drops

Every cow has her own normal. Each cow is scored against her own 14-day history: feeding and chewing changes are lead indicators, while milk yield only lags behind them.



Rumination falls while milk yield stays flat: an early warning.

Illustrative trend. Weights and thresholds are configurable prototype defaults.

COMBINED RISK INDEX

Feed Z-score
+ Rumination Z-score
= Combined Risk Index

14-day	**rolling baseline per cow: her own normal, never a herd average**
80%	**of the risk weight sits on feed time and chewing cycles, the earliest signals**
−30%	**rumination drop with unchanged milk yield triggers instant escalation**

Check Today: only the cows that need a look

Don’t make the farmer monitor every cow. Make the system tell the farmer which cow needs attention first.

EXAMPLE ALERT

Cow #0412

PRIORITY High

PRIMARY CHANGES FLAGGED

Rumination down 35%
Feed time down 22%

SUGGESTED ACTION

Inspect today

Also in the queue: Cow #0187 (High) and Cow #0356 (Medium)

No graph fatigue

Busy staff never scroll through hundreds of tracking plots. All healthy cows are filtered out of the view.

Ranked priority queue

Only anomalous cows appear, ordered by their multi-metric deviation distance so the most urgent is always first.

Reports, not diagnoses

The system reports exact variances, such as “rumination down 35%”, so a human can inspect and diagnose.

Deployable today, smarter tomorrow

The prototype is deliberately focused. These are its current limits and the next improvements.

SCOPE

Prototype boundaries

- ❑ Assumes adequate ambient lighting in the barn
- ❑ Needs camera resolution high enough to capture jaw movement
- ❑ Works on uploaded video batches rather than live streams

ROADMAP

Next: estrus filter

- ❑ A sick cow shows activity down and feed down
- ❑ A cow in heat shows activity up and feed down
- ❑ The filter separates the two before an alert is raised, reducing false alarms

SUMMARY

Conclusion

- ❑ A cost-effective, easily deployable software layer that works with existing farm cameras to maximize herd welfare and preserve dairy yield.
- ❑ Early, objective behavioral data lets farmers and vets act days before milk output falls.

Q&A

Thank you. Questions and discussion welcome.

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